

Infrastructure Progress Report: ExSOIL Containerized Modeling Environment

Period: May–June 2026 **Project:** ExSOIL NSF Prototype **Prepared by:** Steven Wangen, UW–Madison Data Science Institute

Objectives

The ExSOIL project uses the Community Terrestrial Systems Model (CTSM) to simulate soil processes at NEON ecological observatory sites across the United States. Researchers run CTSM at specific tower locations, feed the model observed atmospheric forcing data, and compare the model's predictions of soil temperature, moisture, and carbon fluxes against real measurements from the tower instruments. This model–observation comparison is the foundation for evaluating and improving CTSM's representation of soil biogeochemistry.

The project's computational environment is delivered as a Docker container that packages CTSM, the NEON site workflow, and a Python analysis stack into a single reproducible unit. Researchers pull the container, launch JupyterLab, and work with model simulations and observational data without installing or configuring any of the underlying software.

This reporting period focused on two infrastructure objectives:

1. **Multi-platform support.** The existing container only ran on Intel/AMD processors. Team members on Apple Silicon Macs (M1–M4) could not run the container reliably, limiting local development and testing.
2. **NEON workflow restoration.** The container's NEON tower site workflow had become non-functional due to version incompatibilities in the underlying CESM/CTSM model framework.

Activities and Outcomes

Multi-architecture container rebuild

The project's Docker container was rebuilt from scratch on a modern foundation. The previous container was based on CentOS 8 (end-of-life since December 2021), shipped

Python 3.7 (end-of-life since June 2023), and only produced Intel/AMD binaries.

The rebuilt container:

- **Runs natively on both Intel/AMD and Apple Silicon** (arm64) processors. Docker automatically selects the correct architecture when pulling the image. Team members on Apple Silicon Macs can now develop and test locally without the performance penalties and stability issues of processor emulation.
- **Uses Ubuntu 24.04 LTS** as the base operating system, with security support through 2029. This replaces CentOS 8, which has not received security patches since December 2021.
- **Installs all scientific libraries from conda-forge** pre-built binaries instead of compiling them from Fortran and C source code. This reduced container build time from approximately 45 minutes to 5 minutes and eliminated a major source of architecture-specific build failures.
- **Pins exact package versions** via conda-lock for both processor architectures, ensuring that every team member gets identical software regardless of their hardware.
- **Upgrades to Python 3.13** and current versions of all scientific packages (xarray, cartopy, matplotlib, scipy, etc.), replacing versions that were 4-6 years old.

A 90-test automated validation suite was developed to verify the container's functionality across three tiers: basic environment checks (Python imports, compilers, libraries), CIME case management workflow (creating and configuring CTSM cases), and full Fortran model compilation with scientific analysis workflows (NetCDF I/O, cartopy map rendering, xarray diagnostics). All 90 tests pass on native arm64.

CTSM migration and NEON workflow restoration

During the rebuild, we discovered that the NEON tower site workflow was non-functional. Investigation revealed that NEON support was never part of any CESM 2.x release; it was developed exclusively for standalone CTSM starting in February 2021 and will not enter CESM until version 3.x (currently in beta, targeted for late 2026). The previous container had worked around this by using a custom, undocumented build that mixed component versions from different release lines.

Based on this finding, we replaced the full CESM framework with standalone CTSM 5.4, which is NCAR's intended platform for the NEON workflow. This change:

- **Restored the NEON tower site workflow** with pre-configured settings for 48 NEON sites. Researchers can create and run single-point CLM simulations at any of these sites.

- **Reduced the container image size** by removing approximately 3–4 GB of source code for Earth system components the project does not use (atmosphere, ocean, sea ice, ice sheet, and wave models).
- **Aligned with NCAR's recommended approach** for NEON tower experiments, as documented in the official CTSM user guide and NCAR tutorial materials.
- **Removed the need for several workarounds** that had been applied to make CESM 2.2.2 compile with modern compilers, including PIO2 source patches, Python version pinning, and cmake version restrictions.

The trade-off is that the container no longer supports fully coupled Earth system simulations (where the land surface interacts with a simulated atmosphere, ocean, and ice). The project does not currently use coupled simulations; all existing workflows run CLM in single-point mode with observed atmospheric forcing. If coupled experiments are needed in the future, a separate CESM 3.x container can be maintained alongside the CTSM container.

Documentation

The rebuild and migration are documented through:

- **5 Architecture Decision Records** that capture the rationale for each major technical choice (base image selection, build strategy, dependency management, distributed computing support, and the CTSM migration)
- **A decision brief** analyzing five options for restoring NEON support, including a version lineage diagram and CESM 3.x timeline research
- **A technical rebuild report** covering the full implementation detail for platform engineering review
- **A getting-started guide and introductory notebook** for new users joining the project
- **A roadmap** tracking completed work and planned next steps
- **A changelog** summarizing all additions, changes, and removals

Challenges

The primary challenge was the gap between CESM's official release versions and the NEON workflow's development timeline. The NEON tower site functionality was developed on CTSM's master branch after the CESM 2.2 release line had been cut, and will not be available in a full CESM release until version 3.x ships. The previous container masked this gap by

using an undocumented custom build. Our rebuild exposed the issue, which led to the architectural decision to move to standalone CTSM.

A secondary challenge was achieving Fortran compilation on arm64 with modern compiler and library versions. CESM 2.2.2's code (written primarily for Intel compilers on x86 HPC systems) required 13 separate compatibility fixes to compile with GCC 15 and conda-forge's current NetCDF libraries on ARM processors. These fixes were identified iteratively through the automated test suite. Many of them became unnecessary after the CTSM migration, which uses newer, more portable code.

Next Steps

1. **NEON simulation pipeline.** The Modeling_Hub notebook, which performs model-data evaluation and Kalman Filter calibration, requires CLM history files from a completed transient run and processed NEON tower observations. Building this end-to-end pipeline (data download, simulation execution, observation processing) is the next priority.
2. **CI/CD test integration.** The 90-test validation suite runs locally but is not yet wired into the GitHub Actions workflow. Adding automated testing on both architectures after each push will catch regressions early.
3. **Cross-architecture validation.** All local testing was performed on Apple Silicon (arm64). The Intel/AMD (amd64) image builds in CI but has not been validated through the test suite.
4. **Science notebook validation.** The Data_Hub and Design_Hub notebooks need end-to-end testing with S3 credentials on the new CTSM container to verify the full analysis workflow.